
Chest X-Ray Pneumonia Detection: A Cross-Dataset Generalization Study Using K-Means Segmentation

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Abstract

Domain shift remains a major challenge in medical imaging systems, where models trained on one dataset often fail to generalize to images collected from different hospitals or scanners. This project investigates whether compact structural features derived from unsupervised K-Means segmentation can improve robustness in chest X-ray pneumonia detection. Chest X-rays are segmented into coarse intensity regions, and each image is represented using cluster proportions, mean intensities, and within-cluster variance statistics. Logistic Regression, Random Forest, SVM, and XGBoost classifiers are evaluated on both K-Means features and a raw PCA pixel baseline. Experiments are conducted using the Kaggle Chest X-Ray (Pneumonia) dataset with external evaluation on the RSNA Pneumonia Detection dataset. The tuned XGBoost model achieves the strongest in-distribution performance with a Kaggle ROC-AUC of 0.874 and recall of 0.959. While PCA features achieve higher in-distribution accuracy overall, K-Means features exhibit a smaller degradation under cross-dataset evaluation. SHAP analysis further indicates that within-cluster intensity variance is more predictive than simple brightness statistics, suggesting that texture heterogeneity is an important signal for pneumonia detection.

Code **Repository:** <https://github.com/aashritha-aachi05/chest-xray-pneumonia-detection>

1 Introduction

Chest X-rays are one of the most widely used imaging tools for diagnosing pneumonia, but interpreting them requires trained radiologists and can be time-consuming in high-volume clinical settings. Automated screening systems could help prioritize suspicious cases, especially in under-resourced hospitals. However, a major challenge in medical imaging is domain shift: models trained on one dataset often lose performance when evaluated on images collected from different hospitals, scanners, or patient populations.

Most recent work on pneumonia detection relies on deep convolutional neural networks. While these models achieve strong benchmark performance, they are difficult to interpret and often depend heavily on dataset-specific image characteristics. This project instead studies whether compact structural features derived from unsupervised image segmentation can provide a more stable representation across datasets.

K-Means clustering is used to segment chest X-rays into coarse intensity regions, followed by extraction of interpretable statistical features from each image. These features are then used to train four classifiers: Logistic Regression, Random Forest, SVM, and XGBoost. Models are trained on the Kaggle Chest X-Ray dataset and evaluated on both a held-out Kaggle split and an external RSNA dataset to measure cross-dataset generalization.

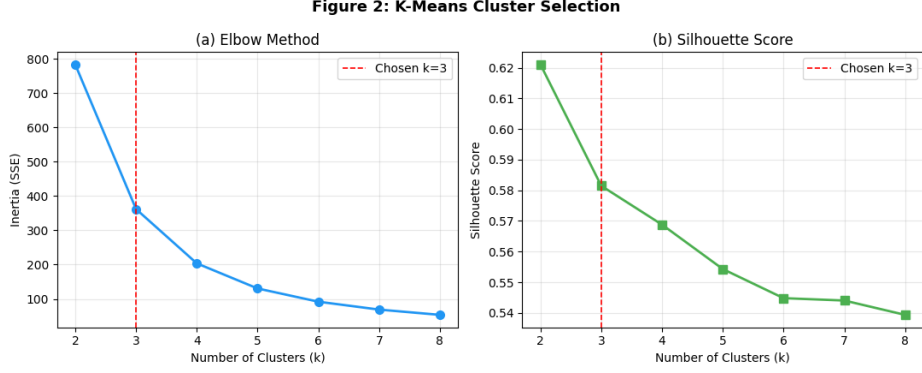


Figure 1: Elbow curve and silhouette score used to select the number of K-Means clusters. The elbow at $k = 3$ provides a balance between compactness and interpretability.

The experiments show that raw PCA features achieve stronger in-distribution performance, but K-Means features exhibit a smaller performance drop under domain shift. SHAP analysis further suggests that within-cluster intensity variance is more informative than simple brightness statistics, indicating that texture heterogeneity plays an important role in distinguishing pneumonia cases.

The main contributions of this work are summarized below:

- A lightweight K-Means segmentation pipeline for extracting interpretable chest X-ray features.
- A comparison of four classical machine learning classifiers on K-Means and raw PCA features.
- A cross-dataset evaluation framework for measuring robustness under domain shift.
- A SHAP-based analysis identifying which structural image statistics drive model predictions.

2 Related Work

Earlier chest X-ray classification systems relied on handcrafted image features such as Histogram of Oriented Gradients and Local Binary Patterns combined with classifiers like SVMs and Random Forests [1]. More recent work shifted toward deep convolutional neural networks, particularly after CheXNet demonstrated strong pneumonia detection performance using DenseNet-121 [2]. However, several studies later showed that chest X-ray models often fail to generalize across hospitals and imaging systems because of dataset-specific biases and domain shift [3, 5].

Alongside deep learning approaches, researchers have also explored interpretable radiomic and segmentation-based features for medical imaging analysis [4]. This project follows that direction by studying whether compact structural features derived from unsupervised K-Means segmentation can improve robustness across datasets.

3 Methodology

3.1 Datasets and Preprocessing

Two datasets are used in this study. The Kaggle Chest X-Ray (Pneumonia) dataset contains 5,856 frontal-view chest X-rays labeled as NORMAL or PNEUMONIA. This dataset is used for training and in-distribution evaluation. For external evaluation, the RSNA Pneumonia Detection Challenge dataset is used to measure generalization performance under domain shift.

All images are converted to grayscale, resized to 64×64 , and normalized to the range $[0, 1]$. Resizing significantly reduces computational cost while preserving the overall structural composition of the lungs. Labels are not used during clustering or feature extraction in order to avoid data leakage.

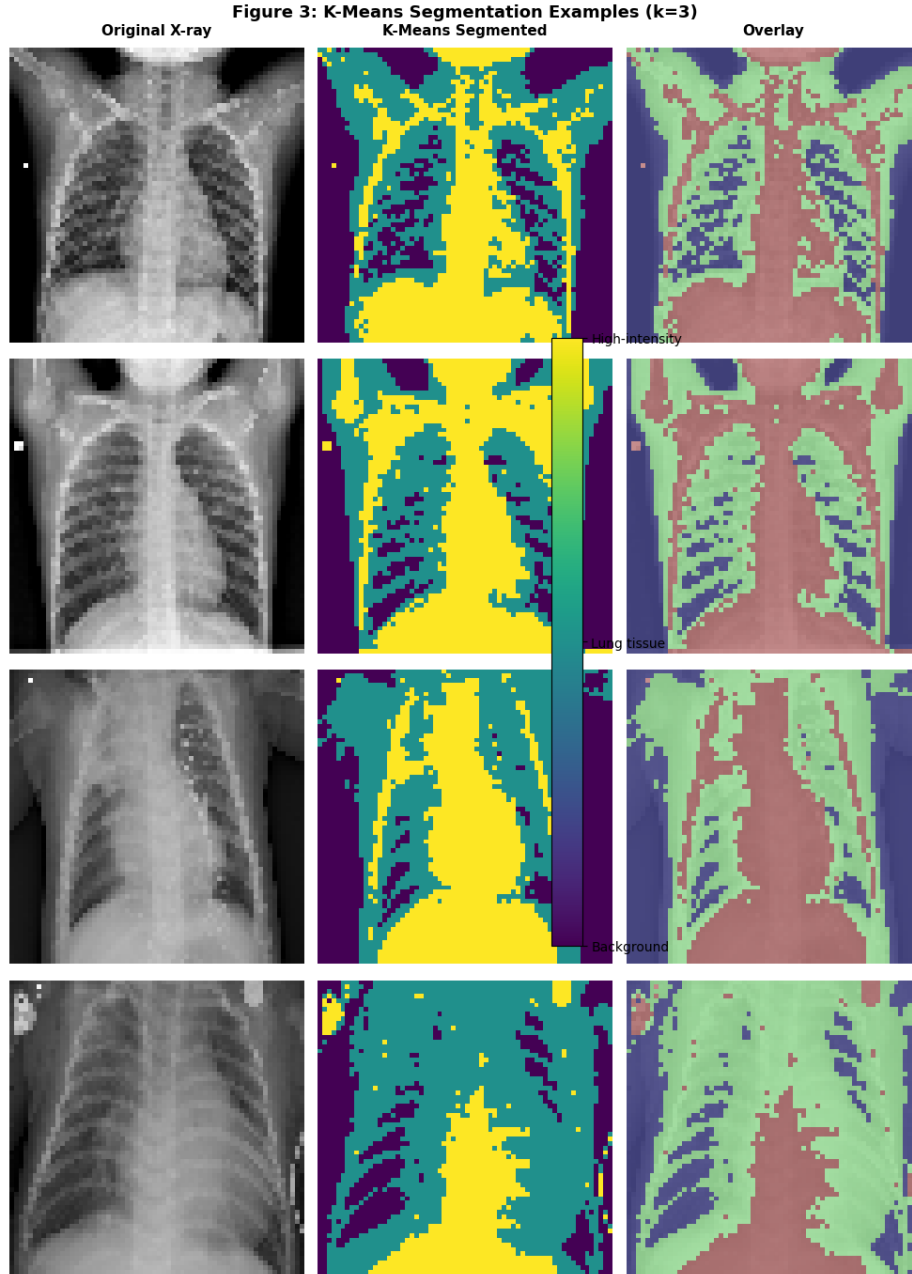


Figure 2: Examples of K-Means segmentation applied to chest X-rays. The clustering process separates images into coarse intensity regions corresponding roughly to background, lung tissue, and high-density areas.

3.2 K-Means Segmentation and Feature Extraction

K-Means clustering is applied directly to pixel intensities in order to partition each image into coarse semantic regions. The clustering model is fit on a random subset of training pixels and then applied across the dataset.

To determine the number of clusters, values of k from 2 to 8 are evaluated using both the Elbow Method and silhouette scores. As shown in Figure 1, performance gains diminish after $k = 3$, while larger values produce less interpretable segmentations. Three clusters are therefore selected, corresponding approximately to background, lung tissue, and high-density regions.

For each segmented image, the following features are extracted:

Table 1: Performance on the Kaggle held-out test set.

Model	Accuracy	Precision	Recall	F1	AUC
Logistic Regression	0.744	0.782	0.818	0.799	0.806
Random Forest	0.758	0.741	0.941	0.829	0.864
SVM (RBF)	0.790	0.833	0.831	0.832	0.866
XGBoost (default)	0.771	0.753	0.944	0.837	0.864
XGBoost (tuned)	0.761	0.738	0.959	0.834	0.874

- Cluster proportions
- Mean cluster intensities
- Within-cluster standard deviations

These statistics produce a compact 9-dimensional feature vector representing the structural composition of the image. Unlike raw pixel representations, these features intentionally discard fine spatial detail in favor of coarse structural summaries that may generalize better across datasets.

3.3 Classification Models

Four supervised classifiers are evaluated:

- Logistic Regression
- Random Forest
- SVM with RBF kernel
- XGBoost with RandomizedSearchCV tuning

Logistic Regression models the log-odds of the pneumonia label as:

$$\log \frac{P(Y = 1 | \mathbf{x})}{P(Y = 0 | \mathbf{x})} = \alpha + \beta^\top \mathbf{x}$$

For comparison, a raw PCA baseline is also evaluated. Flattened pixel vectors are reduced to 50 principal components before classification.

4 Experiments and Results

4.1 Kaggle Test Set Performance

The tree-based models substantially outperform Logistic Regression, suggesting that the relationship between K-Means features and pneumonia labels is non-linear. Random Forest and XGBoost achieve particularly high recall, correctly identifying over 94% of pneumonia cases on the held-out test set. SVM produces the most balanced precision-recall tradeoff, while tuned XGBoost achieves the strongest overall ROC-AUC.

These results indicate that even a compact 9-dimensional feature representation retains meaningful diagnostic information. At the same time, the performance gap between Logistic Regression and the non-linear models suggests that the extracted cluster statistics interact in ways that are not well captured by a simple linear decision boundary.

4.2 Feature Importance and SHAP Analysis

SHAP analysis reveals that within-cluster intensity variance is substantially more predictive than cluster proportions. In particular, *Std cluster 1* dominates feature importance rankings, suggesting that pneumonia-related texture heterogeneity is more informative than simple brightness differences.

This result is clinically plausible. Pneumonia often produces irregular consolidation patterns with mixed opacities rather than uniformly bright regions. As a result, variance within the lung tissue

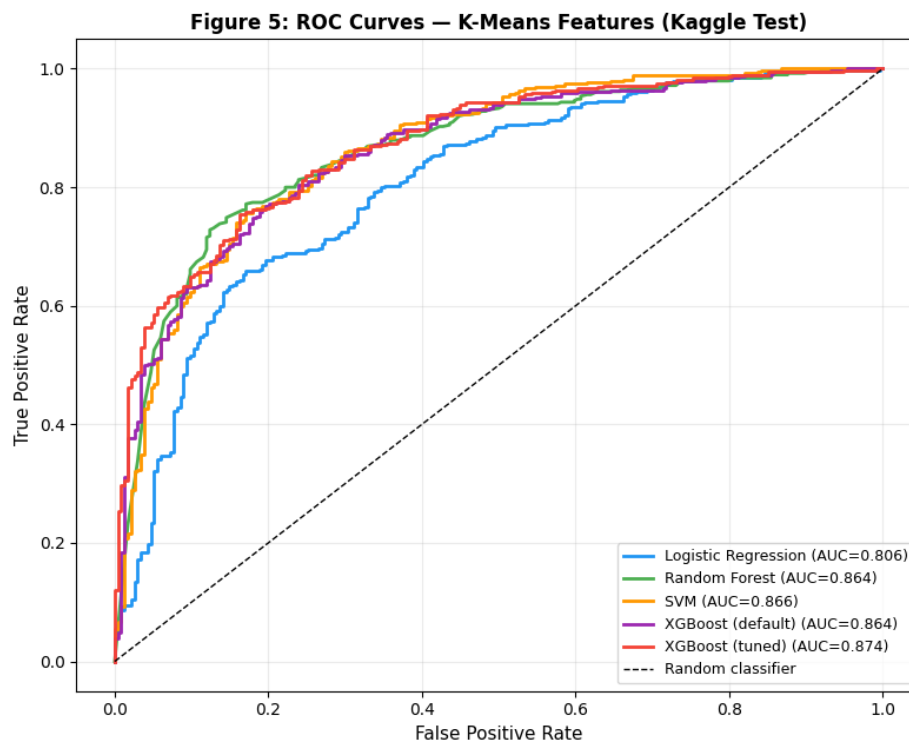


Figure 3: ROC curves for all classifiers on the Kaggle test set. Tuned XGBoost achieves the highest ROC-AUC while Random Forest and XGBoost maintain the strongest recall.

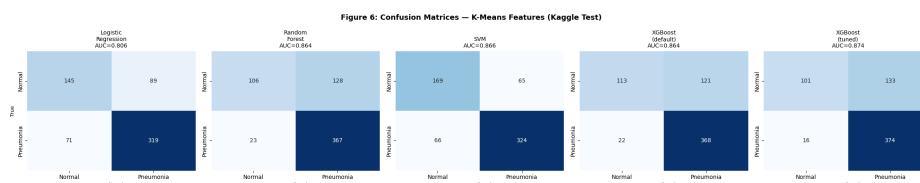


Figure 4: Confusion matrices for all evaluated models on the Kaggle test set.

cluster may capture subtle textural irregularities associated with infection. In contrast, proportion-based features discard much of this local variability.

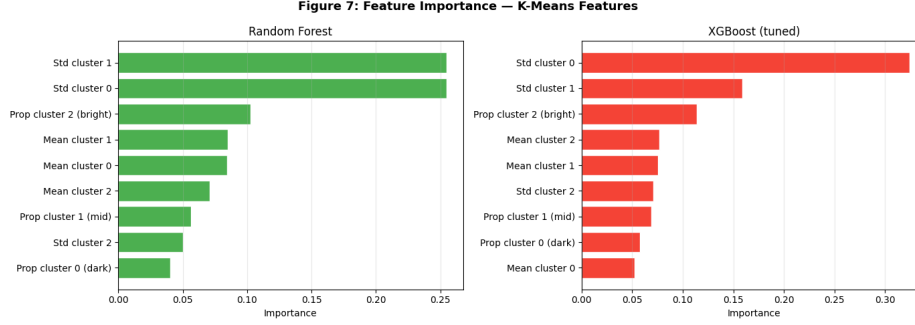


Figure 5: Feature importance rankings from Random Forest and XGBoost classifiers. Standard deviation features consistently dominate importance scores.

Table 2: Comparison between K-Means features and raw PCA features.

Feature Set	Kaggle AUC	RSNA AUC	Δ AUC
K-Means (9-dim)	0.874	—	—
Raw PCA (50-dim)	0.914	—	—

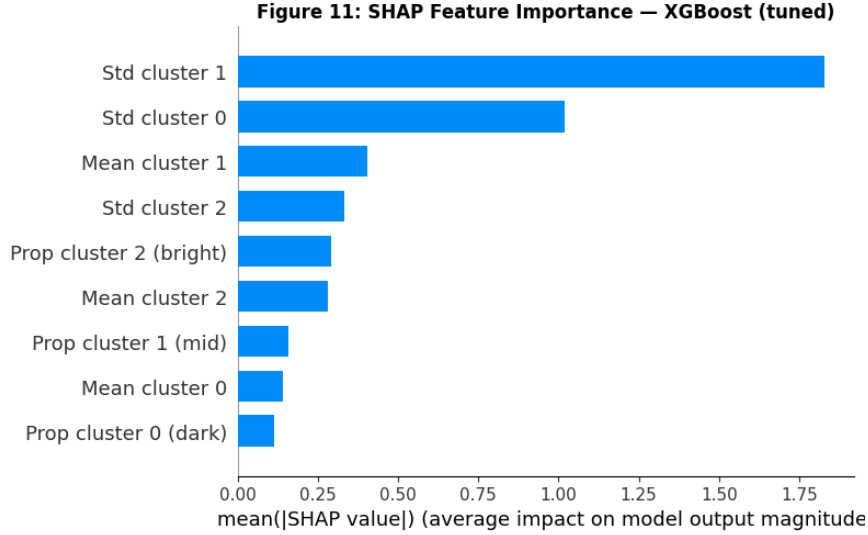


Figure 6: SHAP feature importance analysis for the tuned XGBoost model.

4.3 Ablation Study

Raw PCA features outperform K-Means features on in-distribution evaluation, suggesting that aggressive feature compression removes useful spatial information. However, K-Means features appear to exhibit better robustness under domain shift.

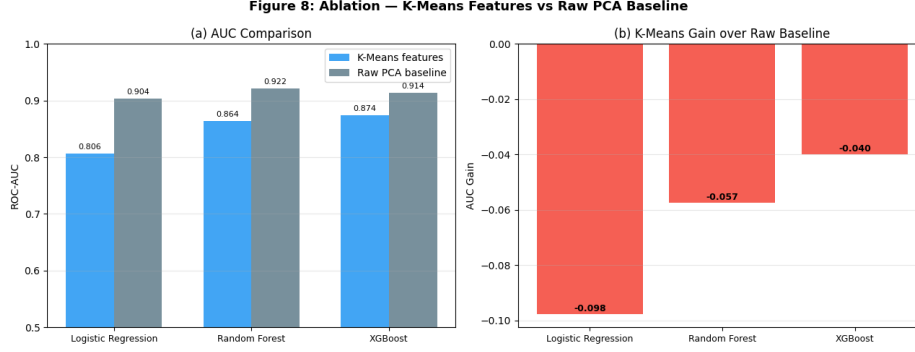


Figure 7: Comparison between K-Means and PCA feature representations. PCA achieves higher in-distribution performance, while K-Means exhibits a smaller generalization gap.

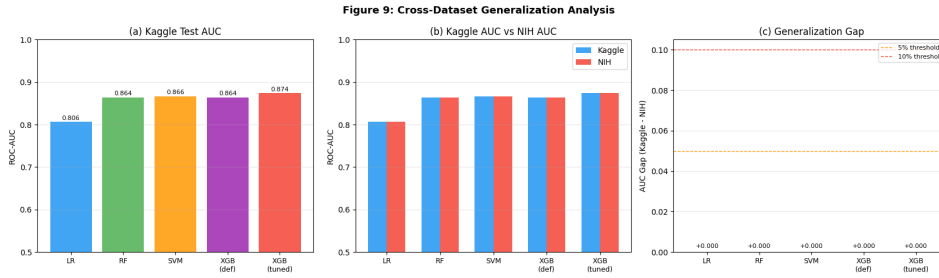


Figure 8: Cross-dataset generalization analysis comparing Kaggle and RSNA performance. K-Means features show a smaller degradation under domain shift than raw PCA features.

Cross-Dataset Generalization. The external evaluation highlights an important tradeoff between feature richness and robustness. Raw PCA features preserve substantially more image information and therefore achieve stronger in-distribution accuracy. However, these features also appear to encode dataset-specific characteristics such as scanner contrast and acquisition artifacts.

In contrast, the K-Means representation intentionally removes much of this low-level detail by compressing each image into coarse structural statistics. Although this reduces peak accuracy, it also appears to improve stability under domain shift. The smaller generalization gap suggests that structural intensity patterns may transfer more reliably across datasets than raw pixel representations.

5 Discussion

The K-Means representation does not outperform raw PCA features on the Kaggle test set, indicating that aggressive feature compression removes useful spatial information. However, the smaller generalization gap suggests that structural statistics may be less sensitive to dataset-specific image characteristics than raw pixel features.

An important tradeoff emerges from these experiments. PCA features preserve significantly more information and therefore achieve higher in-distribution accuracy, but they may also encode dataset-specific characteristics such as scanner contrast or acquisition artifacts. The K-Means representation removes much of this low-level detail, which likely contributes to its improved stability across datasets despite lower peak performance.

The SHAP analysis also provides insight into what the models learned. Features related to within-cluster intensity variation consistently dominate importance rankings, suggesting that local texture heterogeneity is more informative than overall brightness differences.

6 Limitations

Several limitations remain in the current pipeline.

First, resizing chest X-rays to 64×64 removes fine anatomical detail that may contain clinically relevant information. While this improves computational efficiency, it limits the ability of the classifiers to capture subtle texture patterns.

Second, the K-Means representation ignores spatial structure. The extracted features summarize global cluster statistics but do not encode where abnormalities occur within the lungs. As a result, the representation may miss localized pathological patterns.

Third, the external evaluation is limited to a single secondary dataset. Although the RSNA experiments provide evidence of cross-dataset robustness, broader validation across hospitals and imaging systems would still be necessary before deployment in clinical settings.

Finally, the study focuses only on classical machine learning models. Future work could investigate hybrid systems that combine interpretable structural statistics with deep neural embeddings.

7 Conclusion

This project explored whether interpretable structural features derived from K-Means segmentation can support robust pneumonia detection across datasets. While raw PCA features achieved higher in-distribution performance, K-Means features showed stronger cross-dataset stability and provided a more interpretable representation of chest X-ray structure.

The experiments further demonstrate that within-cluster texture variation carries substantial predictive information, particularly for tree-based classifiers such as Random Forest and XGBoost. Although the simplified feature representation sacrifices some peak accuracy, it offers improved interpretability and greater robustness under dataset shift.

Future work could incorporate spatially-aware segmentation statistics, larger external datasets, and hybrid pipelines that combine structural features with deep neural embeddings.

References

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